

UNIX

These lectures teach the UNIX command line, shell scripting, and running analyses on the UCR HPCC cluster. They build on each other, so work through them in order.

	Lecture	Class	Topics
I	Logging In and Command Line Basics (PDF)	Sep 24	the cluster, logging in, OnDemand, paths and navigation, viewing files, getting help, shortcuts, nano, your first shell script
II	Files, Data and Running Programs (PDF)	Sep 29	copying and moving, wildcards, symbolic links, permissions, quotas, redirection and pipes, grep, compression, downloading and transferring data, modules, interactive jobs, Git for homework
III	Shell Programming (PDF)	Oct 1	variables and quoting, conditionals, loops, functions, robust scripts, looping over samples
III lab	Data Processing with Pipes (PDF)	lab / reading	grep, cut, sort/uniq, tr, paste, join, sed, awk; answering questions about GFF and BLAST tables
IV	Running Analyses on the Cluster (PDF)	Oct 29	SLURM batch jobs, choosing resources, job arrays, storage, tmux, conda and containers, find/xargs/parallel

See also the [SSH keys guide](#) and the [Git and GitHub guide](#).



Figure 1: logo